MILESTONE v1a — ARCHITECTURE & ENGINEERING GUIDE

Notes for iPadOS ("Notion with Apple Pencil")

Native Swift 6 • SwiftUI + UIKit/PencilKit • Strict Layered Architecture • 25 Automated Unit Tests

Target: iPadOS 26.5+ (iPad Only) | Repo: github.com/jq-22/notes-app | Release: v1a

1. Executive Summary — What Was Built in Milestone v1a

Milestone v1a delivers a complete, production-structured handwriting notebook app for iPadOS. The codebase is organized into four strict dependency layers (Domain ← Persistence ← Features ← App) so business logic, geometric heuristics, and persistence are 100% unit-testable without UIKit or the simulator.

Notebook → Section → Page

3-column NavigationSplitView with permanent UUIDs, create/rename/reorder/delete (with confirmation), and multi-page batch move/delete.

A4 Paper Layer & Live Thumbnails

595×842 pt A4 portrait canvas. Separate Lined, Squared, and Blank paper background layer + per-page overrides and live ink sidebar thumbnails.

Custom 4-Tool Popover & Ballpoint Pen

Compact top-bar tool menu: Monoline Ballpoint Pen, Highlighter, Rubber (Object/Stroke vs Pixel), and 25%-scaled Ruler toggle + explicit Black/White swatches.

Apple Pencil Double-Tap & Undo/Redo

UIPencilInteraction double-tap toggles between active Pen/Highlighter and Rubber. Top-left Full Screen toggle + Undo and Redo stroke buttons.

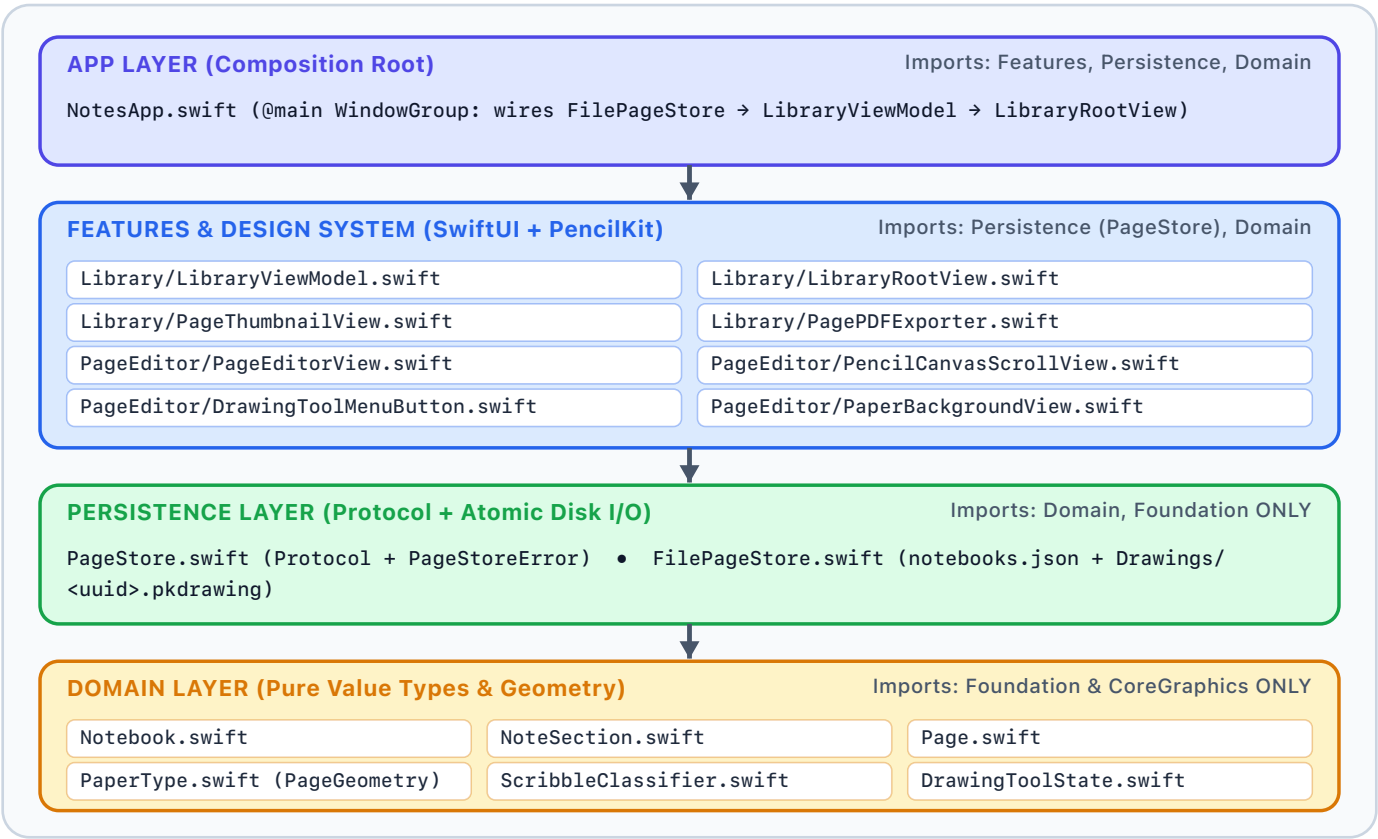
Scribble-to-Erase (1-Step Undo)

Pure geometric classifier (ScribbleClassifier) detects back-and-forth scribbles over existing ink and removes covered strokes in a single undo step.

Fit-Width Zoom, Swipes & PDF Export

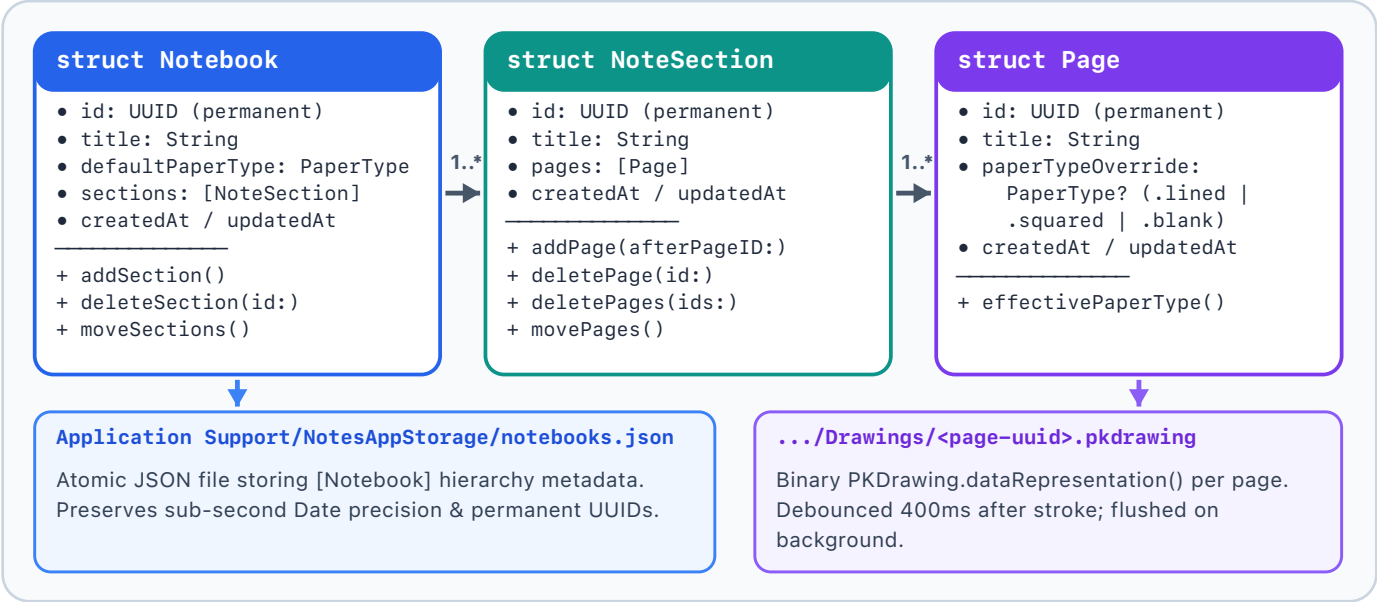
Opens at Fit-to-Width in portrait/landscape, snaps to width on pinch, turns/creates pages on deliberate horizontal drag (0.42s slide), and exports vector PDFs.

2. Diagram 1 — Layered Architecture (Domain ← Persistence ← Features ← App)

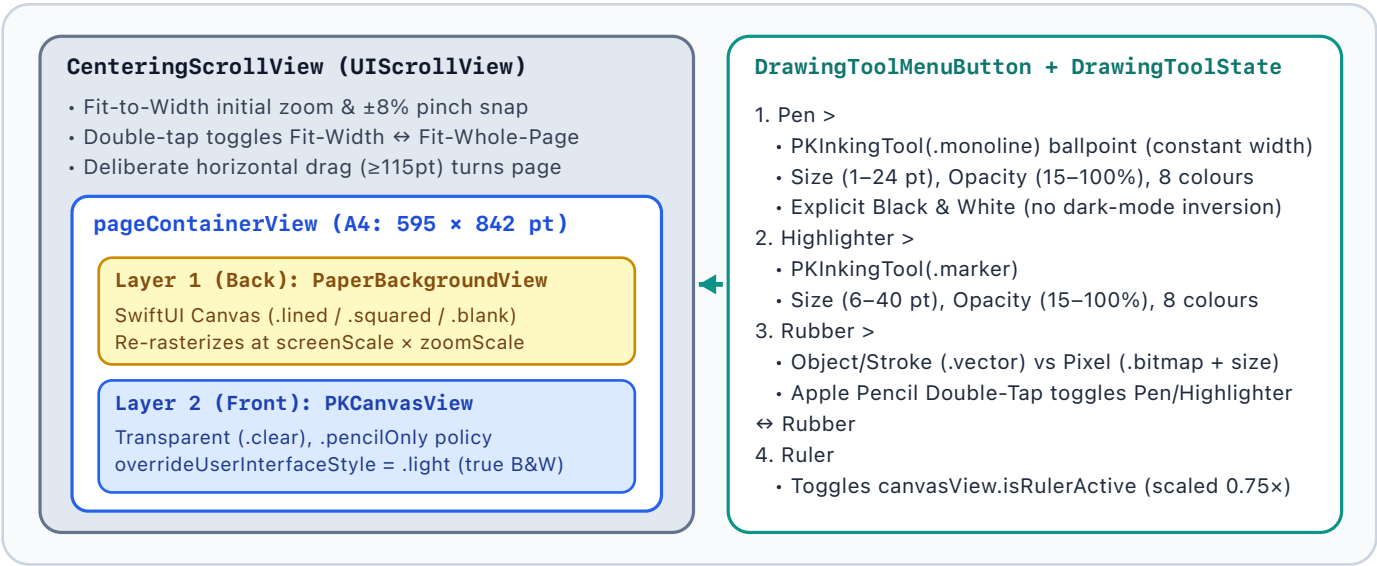


3. Diagram 2 — Domain Data Model (UUID Hierarchy) & Disk Storage Layout

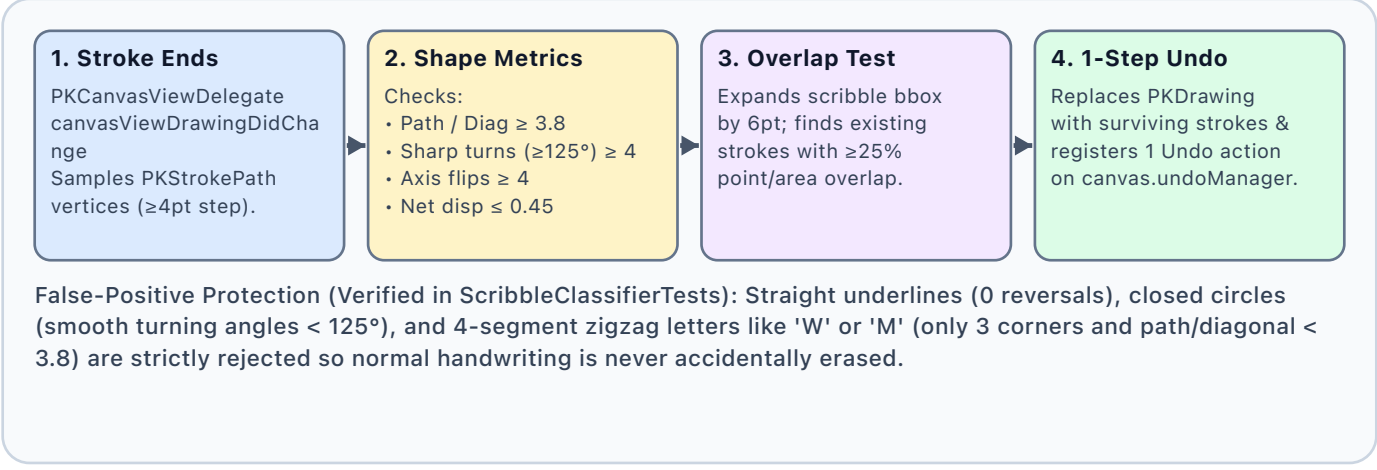
Every Notebook, NoteSection, and Page has a permanent UUID. Nothing is keyed by title or array index, so reordering, moving pages across sections, and future page links/sync never corrupt references. Drawing bytes (PKDrawing) are stored in separate per-page files.



4. Diagram 3 — PageEditor View Stack, Fit-to-Width Zoom & 4-Tool Pipeline



5. Diagram 4 — Scribble-to-Erase Heuristic & Single-Step Undo Pipeline



False-Positive Protection (Verified in ScribbleClassifierTests): Straight underlines (0 reversals), closed circles (smooth turning angles < 125°), and 4-segment zigzag letters like 'W' or 'M' (only 3 corners and path/diagonal < 3.8) are strictly rejected so normal handwriting is never accidentally erased.

6. Complete Codebase File-by-File Reference (What is What)

File Path	Layer	Responsibility & Key Types
Notes/App/NotesApp.swift	App	App entry point (@main). Instantiates FilePageStore and injects it into LibraryViewModel.
Notes/Domain/Notebook.swift	Domain	Root aggregate struct with permanent UUID, title, defaultPaperType, and ordered [NoteSection].
Notes/Domain/NoteSection.swift	Domain	Section value type holding ordered [Page], addPage(afterPageID:), deletePages(ids:), and movePages().
Notes/Domain/Page.swift	Domain	Page metadata struct with permanent UUID, title validation, and optional paperTypeOverride.
Notes/Domain/PaperType.swift	Domain	PaperType enum (.lined, .squared, .blank) and PageGeometry (A4 595×842pt, fit-width & snap math).
Notes/Domain/ScribbleClassifier.swift	Domain	Pure geometric scribble-to-erase classifier with named threshold constants in Configuration.
Notes/Domain/DrawingToolState.swift	Domain	Value type for Pen/Highlighter/Rubber/Ruler, sizes, opacities, Black/White colours, and Pencil double-tap.
Notes/Persistence/PageStore.swift	Persistence	Protocol boundary separating [Notebook] metadata from per-page drawing Data + PageStoreError.
Notes/Persistence/FilePageStore.swift	Persistence	Atomic file persistence in Application Support (notebooks.json + Drawings/<page-uuid>.pkdrawing).
Notes/Features/Library/LibraryViewModel.swift	Features	@Observable @MainActor coordinator: CRUD, selection, 400ms debounced autosave, swipe nav, batch ops.
Notes/Features/Library/LibraryRootView.swift	Features	3-column NavigationSplitView, context menus, rename/delete dialogs, and multi-page selection bar.
Notes/Features/Library/PageThumbnailView.swift	Features	A4 aspect-ratio sidebar thumbnail rendering PaperBackgroundView + live PKDrawing preview.
Notes/Features/Library/PagePDFExporter.swift	Features	Multi-page A4 vector PDF renderer combining paper lines/grid and PKDrawing strokes for sharing.
Notes/Features/PageEditor/PageEditorView.swift	Features	Hosts canvas + top toolbar (Full Screen, Undo/Redo, 4-Tool menu, Scribble toggle, Paper picker).
Notes/Features/PageEditor/PencilCanvasScrollView.swift	Features	UIKit UIScrollView + PKCanvasView: fit/snap width, 0.42s slide page turn, ruler scaling, scribble undo.
Notes/Features/PageEditor/DrawingToolMenuButton.swift	Features	Compact top-bar popover matching sketch: Pen >, Highlighter >, Rubber >, and Ruler toggle.
Notes/Features/PageEditor/PaperBackgroundView.swift	Features	SwiftUI Canvas rendering Lined (with margin), Squared grid, or Blank paper behind the ink layer.
Notes/DesignSystem/Theme.swift	Design	Single source of truth for adaptive colours, spacing, typography, and thumbnail dimensions.

7. Automated Test Suite & Hardware Verification Summary (25 / 25 Passing)

✓ NotebookHierarchyTests (6 tests) — NotesTests/Domain/NotebookHierarchyTests.swift

Verifies paper override resolution, title sanitization, inserting pages below current page, permanent UUID preservation on reorder, safe no-op deletes, and PageGeometry fit/snap-to-width math.

✓ ScribbleClassifierTests (6 tests) — NotesTests/Domain/ScribbleClassifierTests.swift

Synthetic stroke arrays verifying scribble over existing ink triggers erase, while straight lines, closed circles, zigzag 'W'/'M' letters, empty-space scribbles, and disabled toggles never false-trigger.

✓ DrawingToolStateTests (3 tests) — NotesTests/Domain/DrawingToolStateTests.swift

Verifies Apple Pencil double-tap toggling between Pen/Highlighter and Rubber, stroke width/opacity clamping, and independent Ruler toggling.

✓ FilePageStoreTests (4 tests) — NotesTests/Persistence/FilePageStoreTests.swift

Verifies empty store initialization, full Notebook JSON round-trip with sub-second Date precision, isolated per-page .pkdrawing lifecycle, and corrupted JSON error surfacing.

✓ LibraryViewModelTests (6 tests) — NotesTests/Features/LibraryViewModelTests.swift

Verifies full hierarchy creation & relaunch restore, drawing cleanup on delete, error alert surfacing, 400ms debounced autosave & page-switch flush, swipe page navigation, and multi-page batch move/delete/PDF export.